Week 1 Design Document: MLFQ Scheduler in xv6-RISC-V

Project: Multi-Level Feedback Queue (MLFQ) Scheduler
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1. Introduction

The goal of this project is to replace xv6's default round-robin scheduler with a Multi-Level Feedback Queue (MLFQ) scheduler. This allows dynamic priority scheduling based on process behavior, improving responsiveness for I/O-bound processes while preventing starvation of CPU-bound processes.

2. System Call: getprocinfo

- **Purpose:** Retrieve pid, state, and MLFQ queue for debugging.
- **Implementation:** Added in sysproc.c with kernel-side logic in proc.c.
- **Testing:** User program calls getprocinfo and prints process states to verify correctness, using user systemcall 'pinfo'.

3. Scheduler Design

3.1 Queue Structure

- **Number of Queues:** 4 priority levels (Q0–Q3)
- **Scheduling Policy:** Round-robin within each queue
- **Queue Promotion/Demotion:**
  - Demotion: Process uses up its timeslice → moves to lower-priority queue
  - Promotion: I/O-bound or yielding process → moves to higher-priority queue
- Circular buffers implemented via struct proc_queue.

3.2 Time Quanta

- Q0 (highest priority): 4 ticks
- Q1: 8 ticks
- Q2: 16 ticks
- Q3 (lowest priority): ∞ ticks (infinite)

3.3 Process Tracking

- Each struct proc extended to include:
  - int queue
  - int time_in_queue ticks spent in this queue
  - int in_queue flag to prevent double enqueue

3.4 Queue Scaffolding

Implemented:

- enqueue()
- dequeue()
- is_empty()
- print_queue()
  All with locking. Queues initialized in procinit()

3.5 Demotion/Promotion Policy

- **Demotion:** After exhausting timeslice without yielding, process moves down one level.
- **Promotion:** To prevent starvation and ensure fairness, after a fixed boost interval, all runnable processes are moved back to the highest-priority queue, allowing them to compete for CPU fairly again. (starvation prevention is planned in Week 3).

4. Testing

- getprocinfo verified
- queue insertion validated through print_queue()
- timer tick increments confirmed

5. Future Work

- Implement full MLFQ logic in proc.c with demotion/promotion
- Time-slice enforcement and priority adjustment
- Starvation prevention with priority boosting
- Extend getprocinfo to display queue-level transitions and timeslice usage